Demand, Supply, and Labor Market Shocks: A Multivariate Analysis of U.S. Inflation (1992–2024)

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1. Introduction

Understanding the drivers of inflation remains one of the central questions in macroeconomics. Inflation is not determined in isolation. It emerges from dynamic interactions among real economic activity, consumer demand, and labor market conditions. Industrial production reflects supply-side output fluctuations, retail sales capture demand-side spending pressures, and unemployment measures labor market slack. These variables evolve jointly over time and influence each other through complex feedback mechanisms. A modeling framework that treats them independently may overlook important transmission channels.

Traditional univariate time series models, such as ARIMA, focus on forecasting a single variable based on its own past behavior. While useful for prediction, these models do not allow for simultaneous interactions among multiple macroeconomic variables. Even dynamic regression approaches often impose an implicit assumption of exogeneity, treating some variables as purely explanatory. In macroeconomic systems, however, such assumptions are often unrealistic, as inflation can influence unemployment, unemployment can affect consumer demand, and output dynamics can feed back into prices.

To address these limitations, this study begins with a Vector Autoregression (VAR) framework as a baseline, a model introduced by Sims (1980) that treats all variables as endogenous and allows for dynamic feedback without imposing strong structural restrictions. Preliminary analysis of the data reveals that all four series are integrated of order one, I(1), meaning they contain unit roots. When I(1) series share common long-run equilibrium relationships, a property called cointegration, estimating a VAR on first differences discards meaningful long-run information. The Johansen (1991) trace test identifies three cointegrating relationships among the variables, motivating a Vector Error Correction Model (VECM) as the preferred specification. The VECM extends the VAR framework by incorporating error correction terms that capture the speed and direction of adjustment back to long-run equilibria.

The empirical analysis uses monthly U.S. data from January 1992 through October 2024 (407 observations). The dataset includes the Consumer Price Index for All Urban Consumers (CPIAUCSL), Industrial Production (INDPRO), Advance Retail Sales (RSXFS), and the Unemployment Rate (UNRATE). Together, these variables form a multivariate macroeconomic system capturing demand-side, supply-side, and labor market influences on inflation.

The central research question of this paper is the following:

> **How do demand, supply, and labor market shocks dynamically affect U.S. inflation over time?**

By estimating a VECM and analyzing impulse response functions and forecast error variance decompositions, this study quantifies the transmission mechanisms through which shocks to real economic

activity propagate to prices, both in the short run and over the long run.

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2. Methodology

2.1 Stationarity and Integration

A time series y_t is said to be integrated of order one, denoted I(1), if it is non-stationary in levels but stationary after first differencing. The Augmented Dickey-Fuller (ADF) test is used to test the null hypothesis of a unit root. Failure to reject the null implies the series is non-stationary. Rejection after differencing confirms I(1) status. Establishing the order of integration for each series is the first step in the modeling process, as it determines whether standard regression methods apply or whether cointegration analysis is required.

2.2 The Vector Autoregression Framework

Let Y_t denote a $k \times 1$ vector of time series variables. A Vector Autoregression of order p, denoted VAR(p), is defined as:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + \varepsilon_t$$

where c is a $k \times 1$ vector of intercepts, A_i are $k \times k$ coefficient matrices, and $\varepsilon_t \sim (0, \Sigma)$ is a vector of white noise innovations with covariance matrix Σ.

Unlike univariate AR models, the VAR framework treats all variables as endogenous. Each variable depends on its own lags and the lags of all other variables in the system, allowing for dynamic feedback without imposing strong theoretical restrictions. The VAR requires all variables to be stationary, however. When variables are I(1), the standard practice is to difference them before estimation, but this approach discards long-run equilibrium relationships if cointegration is present.

2.3 Cointegration and the VECM Framework

When two or more I(1) series share a common stochastic trend, they are said to be cointegrated. Cointegrated series drift together over time such that certain linear combinations of them are stationary. Granger and Engle (1987) showed that if cointegration exists, a VAR on first differences is misspecified, omitting the error correction mechanism that governs how the system adjusts back toward long-run equilibrium.

The Vector Error Correction Model (VECM) resolves this by decomposing the dynamics into a long-run component and a short-run adjustment component:

$$\Delta Y_t = \alpha \beta' Y_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t$$

where:

- β is a $k \times r$ matrix of **cointegrating vectors** that define the r long-run equilibrium relationships ($\beta' Y_{t-1}$ are the error correction terms, ECTs)
- α is a $k \times r$ matrix of **adjustment coefficients** measuring the speed at which each variable corrects toward equilibrium
- Γ_i are $k \times k$ matrices of **short-run dynamic coefficients** capturing transitory interactions among first-differenced variables
- r is the **cointegrating rank**, estimated via the Johansen trace test

The error correction terms $\beta' Y_{t-1}$ represent deviations from the long-run equilibria. When a variable drifts above its equilibrium, a negative and significant α coefficient pulls it back. This error-correcting behavior is what distinguishes the VECM from a VAR on differences and enables identification of long-run relationships that would otherwise be lost.

2.4 Lag Length and Rank Selection

Lag length for the underlying VAR is selected using information criteria, specifically the Bayesian Information Criterion (BIC/SC), which penalizes model complexity more heavily than AIC and is appropriate for monthly macroeconomic data. The lag order from VARselect determines the $k - 1$ lags entering the VECM in differences.

The cointegrating rank r is determined using the Johansen trace test. The test sequentially evaluates the null hypothesis that the number of cointegrating relationships is at most r, beginning with $r = 0$. The test statistic is compared against critical values, and the rank is set to the value at which the null first fails to be rejected.

2.5 Impulse Response Functions

Impulse Response Functions (IRFs) trace the dynamic effect of a one-standard-deviation shock to one variable on the current and future values of all variables in the system. Because the residuals ε_t may be contemporaneously correlated, shocks must be orthogonalized. This study employs a Cholesky decomposition of the covariance matrix $\Sigma = P P'$, where P is lower triangular, yielding orthogonal structural innovations u_t such that $\varepsilon_t = P u_t$.

In the VECM context, IRFs are computed from the VAR representation of the estimated VECM, making them directly comparable to VAR IRFs but reflecting the system's cointegrated structure. IRFs from a VECM may exhibit persistent, non-zero long-run responses, precisely because the system converges to cointegrating equilibria rather than the origin.

Bootstrap confidence intervals (95%, 100 runs) are used to assess the statistical significance of impulse responses.

2.6 Forecast Error Variance Decomposition

The Forecast Error Variance Decomposition (FEVD) partitions the h-step-ahead forecast error variance of each variable into proportions attributable to shocks in each variable of the system. This provides insight into the relative importance of different transmission channels at various horizons, distinguishing between short-run (own-shock driven) and long-run (cross-variable) dynamics.

2.7 Toy Example: Simulated VECM

To illustrate the mechanics of a VECM and the interpretation of error correction terms, we construct a simple simulated bivariate cointegrated system. Let x_t follow a random walk:

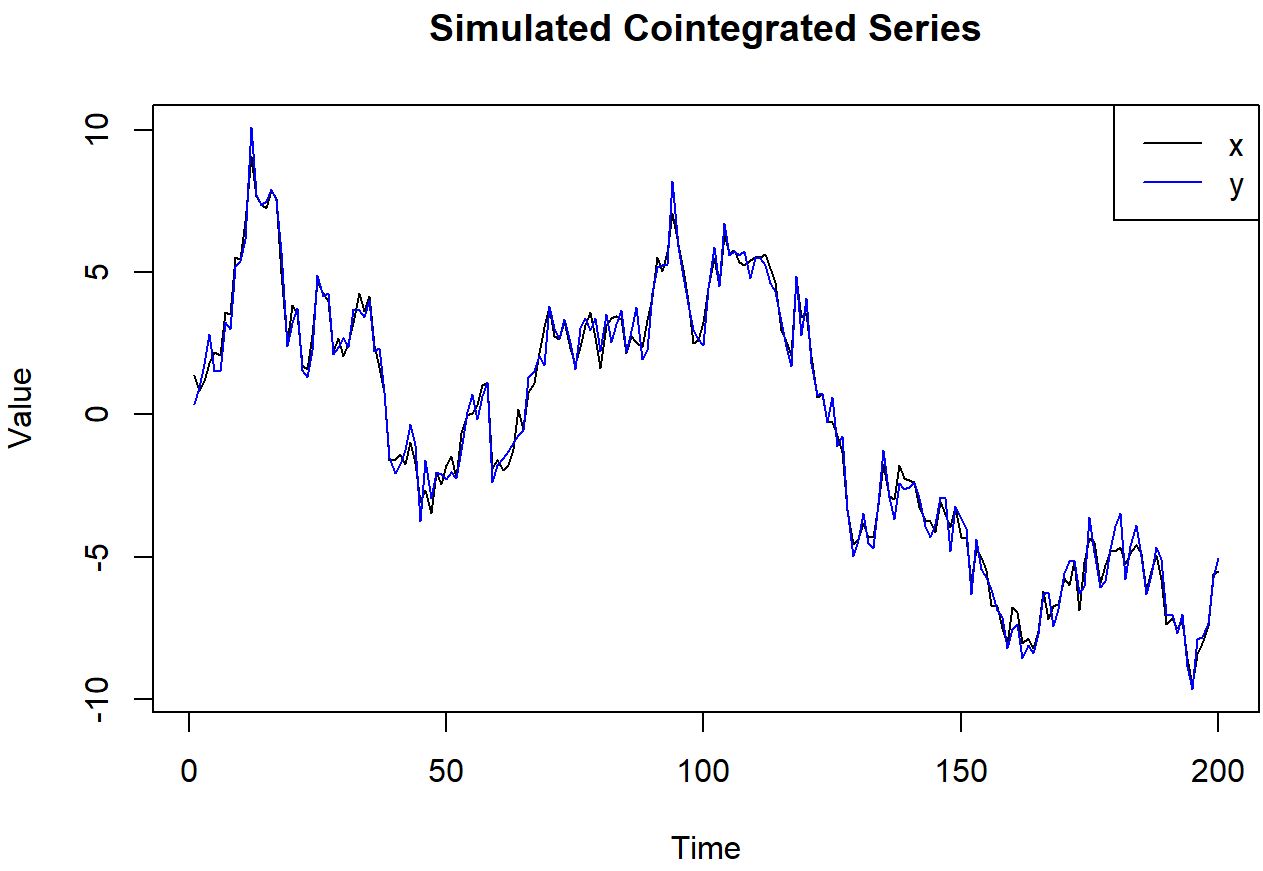
$$x_t = x_{t-1} + u_t, \quad u_t \sim N(0, 1)$$

and let y_t be cointegrated with x_t:

$$y_t = x_t + \eta_t, \quad \eta_t \sim N(0, 0.5^2)$$

By construction, $y_t - x_t \sim N(0, 0.25)$ is stationary, so the pair (y_t, x_t) is cointegrated with cointegrating vector $(1, -1)'$.

The Johansen trace test applied to simulated data from this system confirms a single cointegrating relationship ($r = 1$). Estimating a VECM with $r = 1$ recovers an error correction term that is negative and statistically significant in the y equation, meaning y corrects back toward x when the two diverge, and near zero in the x equation, correctly identifying x as the driver that does not self-correct. This asymmetry is the defining feature of cointegrated systems and motivates the VECM's use in the empirical application.



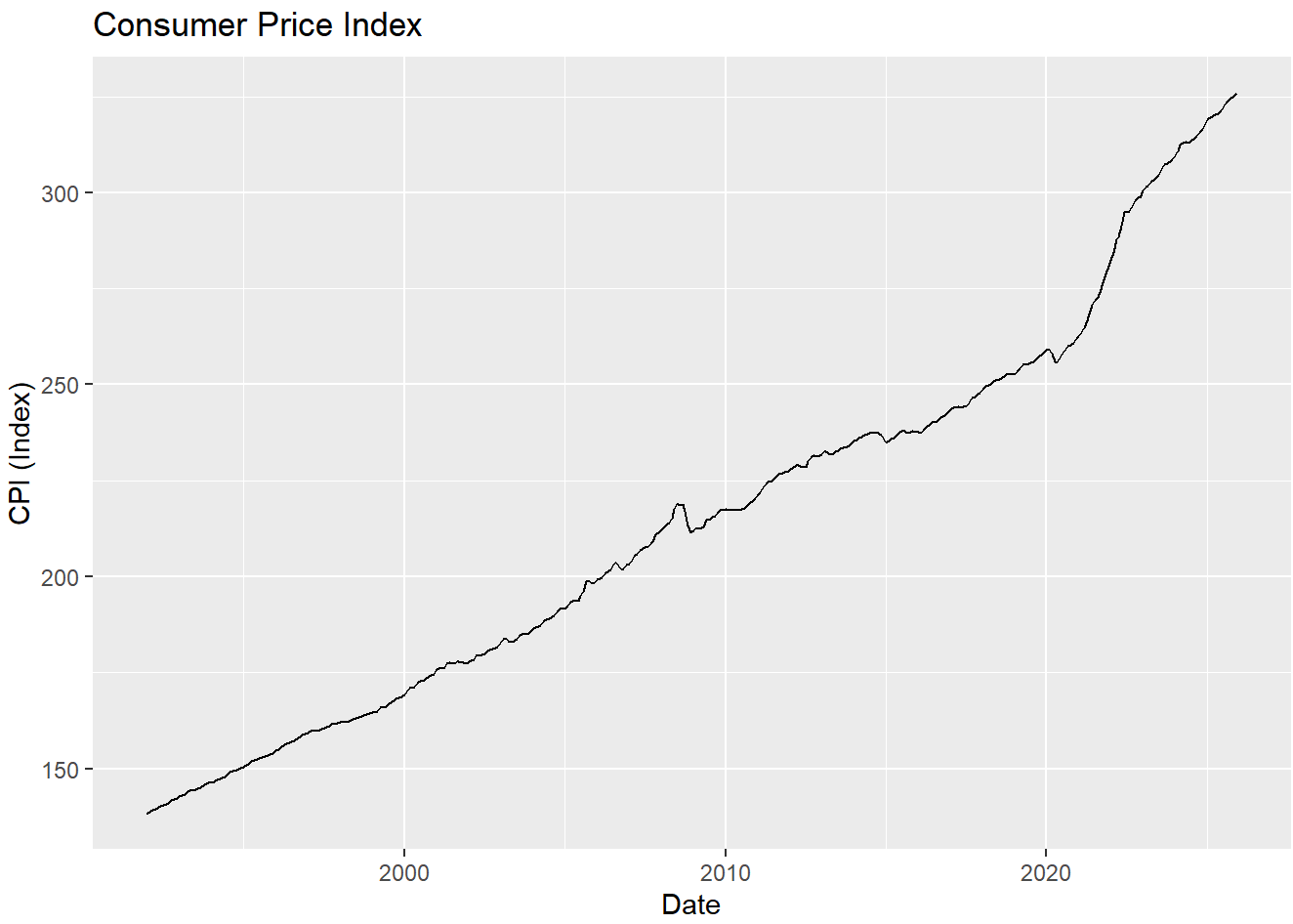
3. Empirical Application

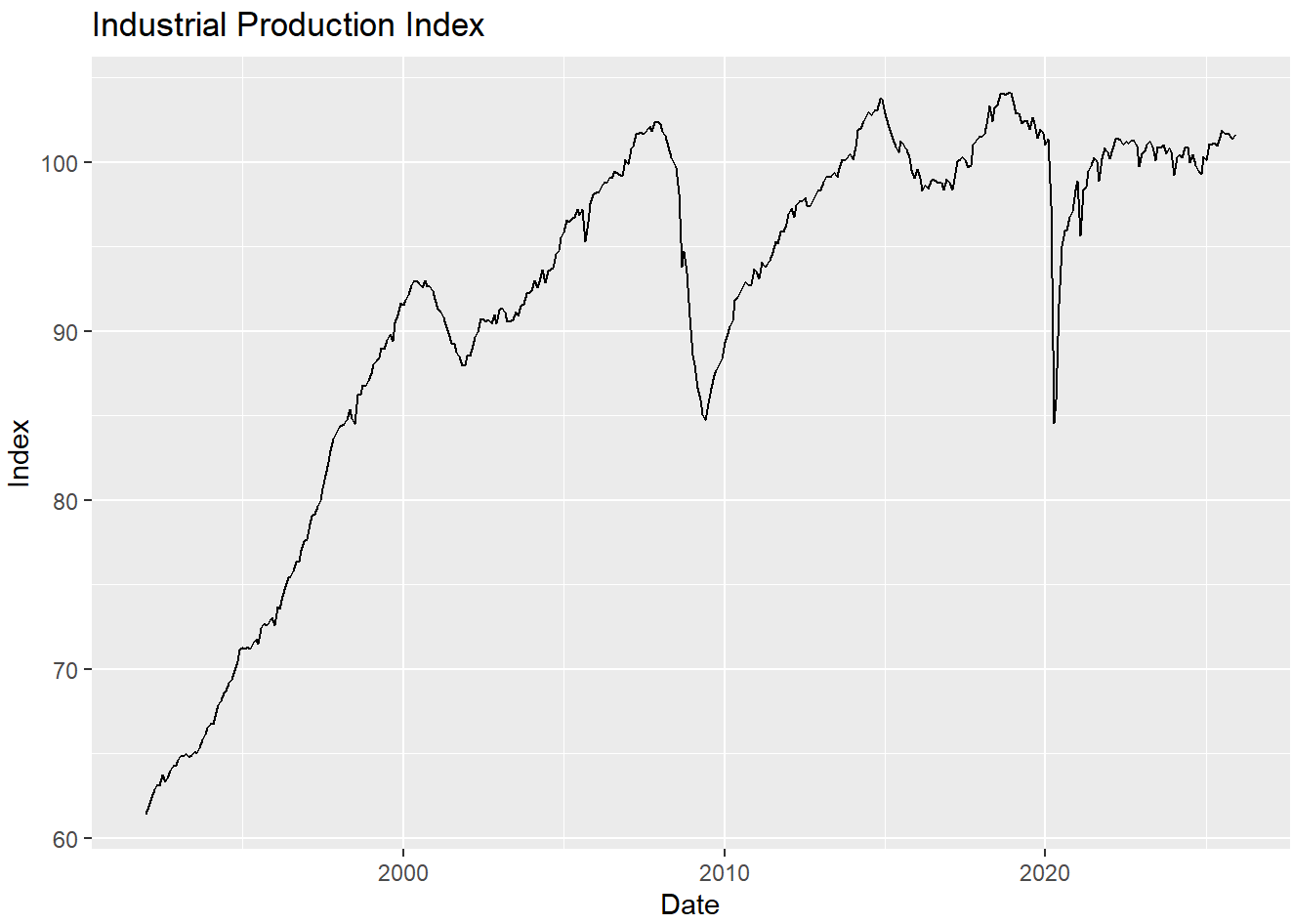
3.1 Data Description

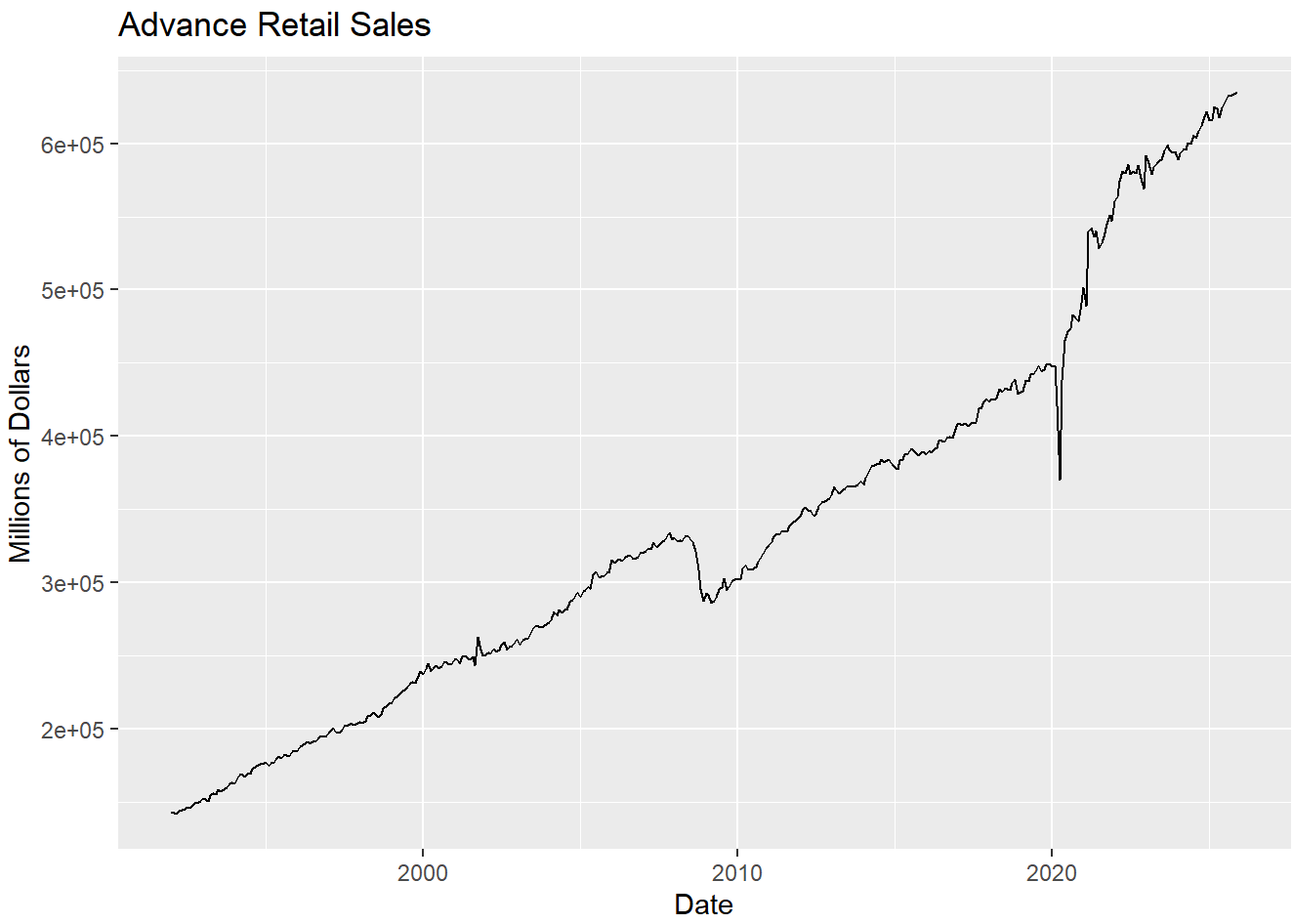
The empirical analysis uses monthly U.S. macroeconomic data from January 1992 through October 2024 (407 observations). The dataset includes four key variables obtained from the Federal Reserve Economic Data (FRED) database, all seasonally adjusted:

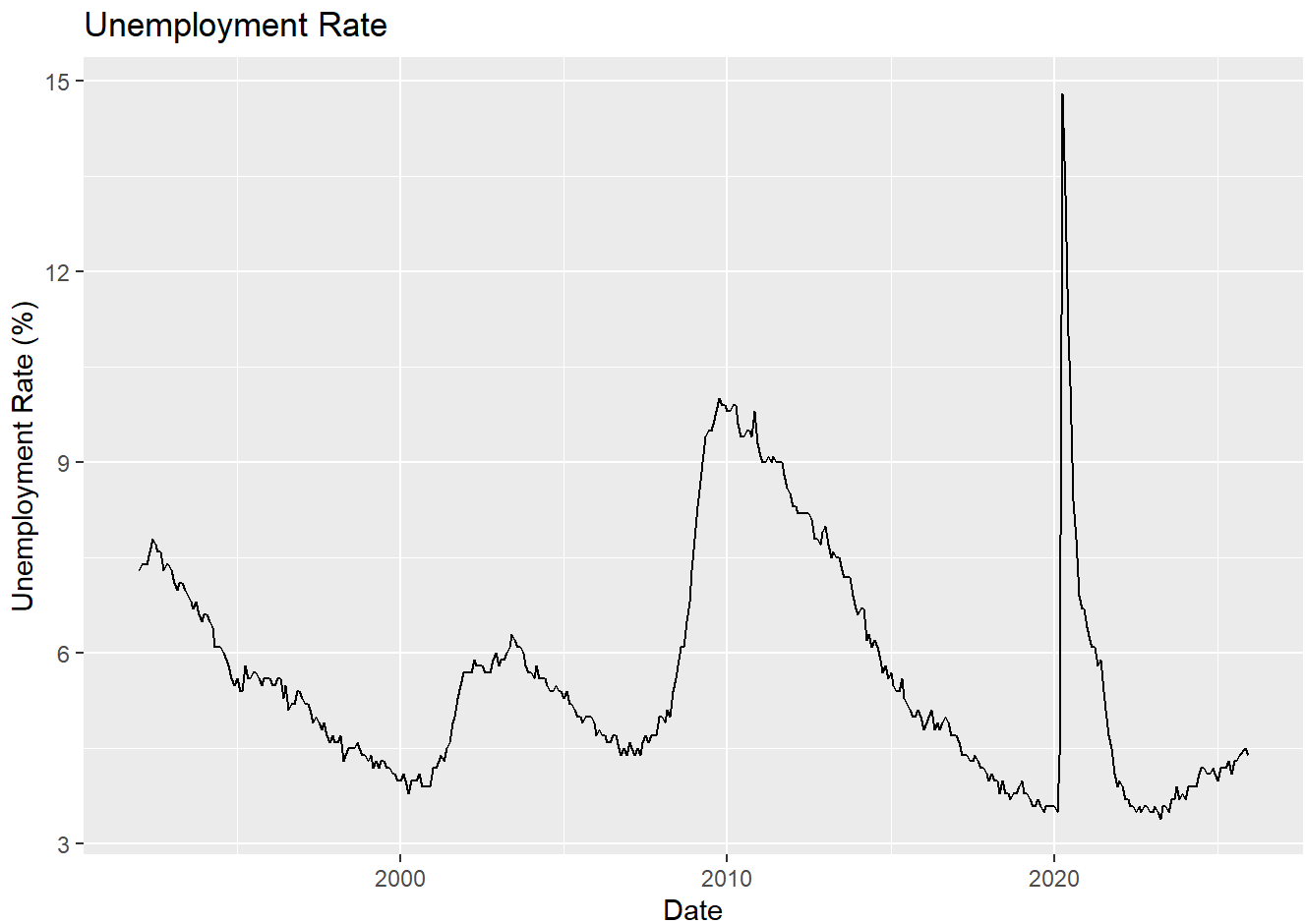
1. **Consumer Price Index (CPIAUCSL)**, used as the primary measure of the price level
2. **Industrial Production (INDPRO)**, which captures aggregate real output in manufacturing, mining, and utilities and serves as a proxy for supply-side conditions
3. **Advance Retail Sales (RSXFS)**, which serves as a proxy for consumer demand
4. **Unemployment Rate (UNRATE)**, which reflects labor market slack

Together, these series form a multivariate macroeconomic system designed to capture demand-side, supply-side, and labor market influences on inflation over more than three decades of U.S. economic history.


Consumer Price Index
CPI (Index)
300
250
200
150
2000
2010
2020
Date


Industrial Production Index
Index
100
90
80
70
60
2000
2010
2020
Date


Advance Retail Sales
Millions of Dollars
6e+05
5e+05
4e+05
3e+05
2e+05
2000
2010
2020
Date



3.2 Stationarity Testing

Before model selection, the order of integration for each series is tested using the Augmented Dickey-Fuller (ADF) test. The null hypothesis is the presence of a unit root.

In levels, all four series fail to reject the null hypothesis at conventional significance levels ($p > 0.05$), confirming that CPI, industrial production, retail sales, and the unemployment rate are each non-stationary. After first differencing, all four series reject the null ($p < 0.05$), establishing that each is integrated of order one, I(1).

This result rules out estimation of a VAR in levels and signals that the appropriate next step is to test for cointegration rather than simply differencing all variables.

3.3 Cointegration Testing

Given that all four series are I(1), the Johansen trace test is applied to determine the number of cointegrating relationships r. The test is conducted using the lag order selected by BIC from VARselect.

Null Hypothesis	Test Statistic	10% Critical Value	Decision
$r = 0$	145.97	60.16	Reject
$r \leq 1$	53.41	41.07	Reject
$r \leq 2$	25.14	24.60	Reject
$r \leq 3$	7.03	12.97	Fail to Reject

The trace test strongly rejects the null of no cointegration and sequentially rejects up through $r \leq 2$. The null $r \leq 3$ is not rejected, indicating that three cointegrating relationships exist among the four variables ($r = 3$).

Because $r > 0$, a VAR estimated on first differences would discard the long-run equilibrium information encoded in the cointegrating vectors. The VECM is therefore the appropriate specification.

3.4 Model Selection: VAR vs. VECM

The decision between VAR and VECM follows directly from the cointegration test results. The general decision rule is:

- If all series are stationary (I(0)): estimate VAR in levels
- If series are I(1) with $r = 0$ (no cointegration): estimate VAR in first differences
- If series are I(1) with $r > 0$ (cointegration present): estimate VECM

Because the Johansen test identifies $r = 3$, the VECM with three error correction terms is selected as the primary model. The VAR framework served as a useful diagnostic baseline, motivating the stationarity testing and cointegration analysis that revealed the richer long-run structure, but the VECM is the preferred specification for inference.

3.5 VECM Results: CPI Equation

The VECM is estimated with $r = 3$ cointegrating vectors. The CPI equation, governing changes in the Consumer Price Index, is the primary focus given the paper's research question.

Key results from the CPI equation:

- **R^2 = 0.587**: the model explains approximately 58% of month-to-month variation in the CPI, indicating strong explanatory power for a first-differenced macroeconomic specification
- **ect1 = −0.030** ($p < 0.01$): the first error correction term is negative and highly significant, confirming that when CPI is above its long-run equilibrium with the other variables, it corrects downward at approximately 3% per month, a meaningful and statistically robust speed of adjustment
- **cpi.dl1 = 0.317** ($p < 0.001$): lagged CPI growth is strongly positive, capturing significant short-run inflation momentum

- **retail_sales.dl1 = 2.18e−05** ($p < 0.001$): lagged retail sales growth is highly significant, indicating that demand-side shocks transmit meaningfully to inflation even in the short run
- **ind_pro.dl1**: the supply shock is statistically insignificant in the contemporaneous CPI equation, suggesting supply-side effects on inflation operate over longer horizons rather than within a single month
- **unemp_rate.dl1**: the labor market shock is also insignificant in the CPI equation at the one-month lag, consistent with labor market effects on inflation being mediated through long-run equilibrium dynamics captured by the ECT

3.6 Model Stability

Model stability is assessed by examining the characteristic roots of the estimated system. For a VECM with $r = 3$ cointegrating relationships among $k = 4$ variables, there should be exactly $r = 3$ unit roots and $k - r = 1$ non-explosive root in the companion VAR representation. All remaining moduli lie strictly inside the unit circle, confirming that the estimated VECM is dynamically stable and that impulse responses and variance decompositions are well-defined.

3.7 Diagnostic Tests

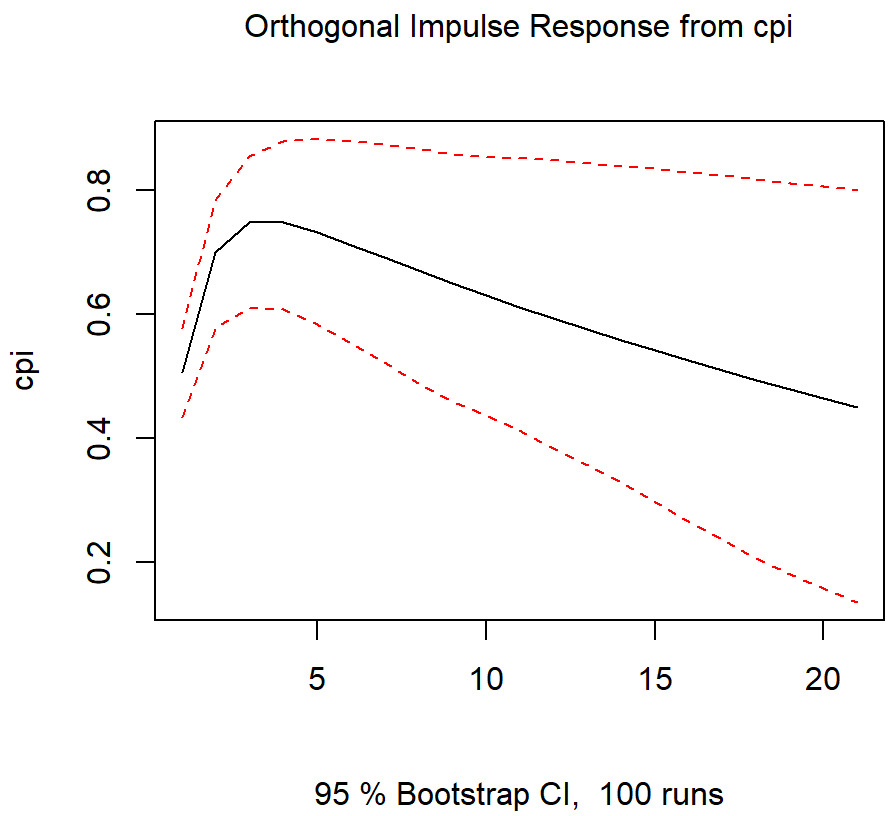
Three standard residual diagnostic tests are applied to the estimated VECM, converted to its VAR representation for testing:

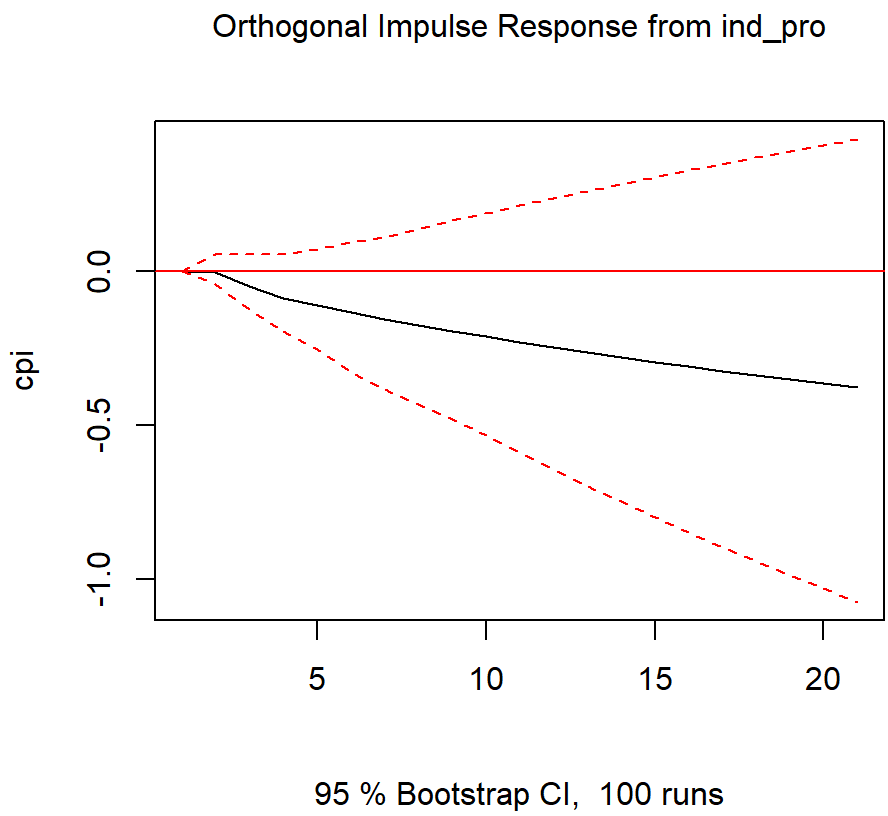
- **Serial Correlation (Portmanteau test)**: rejects the null of no serial correlation
- **Heteroscedasticity (ARCH-LM test)**: rejects the null of constant variance
- **Normality (Jarque-Bera test)**: rejects the null of multivariate normality

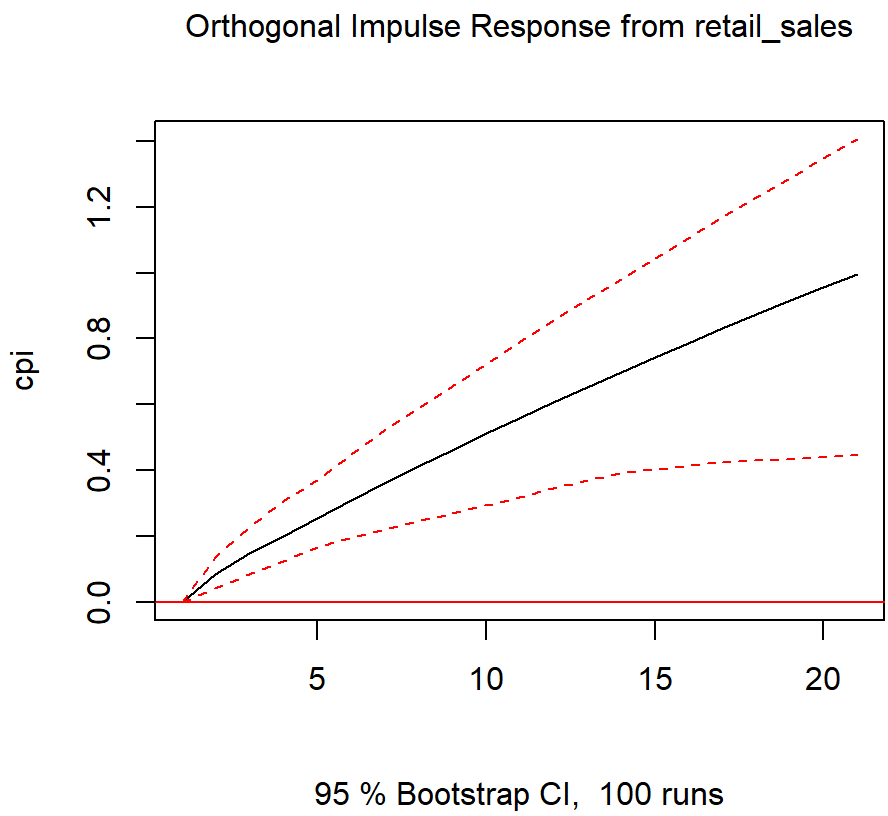
All three diagnostics reject their null hypotheses. The primary driver of these violations is the COVID-19 period (2020–2022), which introduced extreme observations and structural breaks that are difficult for a linear, constant-coefficient model to absorb. Despite these violations, the VECM estimates, particularly the cointegrating relationships, error correction coefficients, and long-run IRF patterns, remain interpretable as approximations of the underlying macroeconomic dynamics. Confidence bands from the bootstrap procedure should be treated with appropriate caution. One natural extension would be to incorporate a structural break dummy for 2020–2022 or restrict the sample to the post-2010 period.

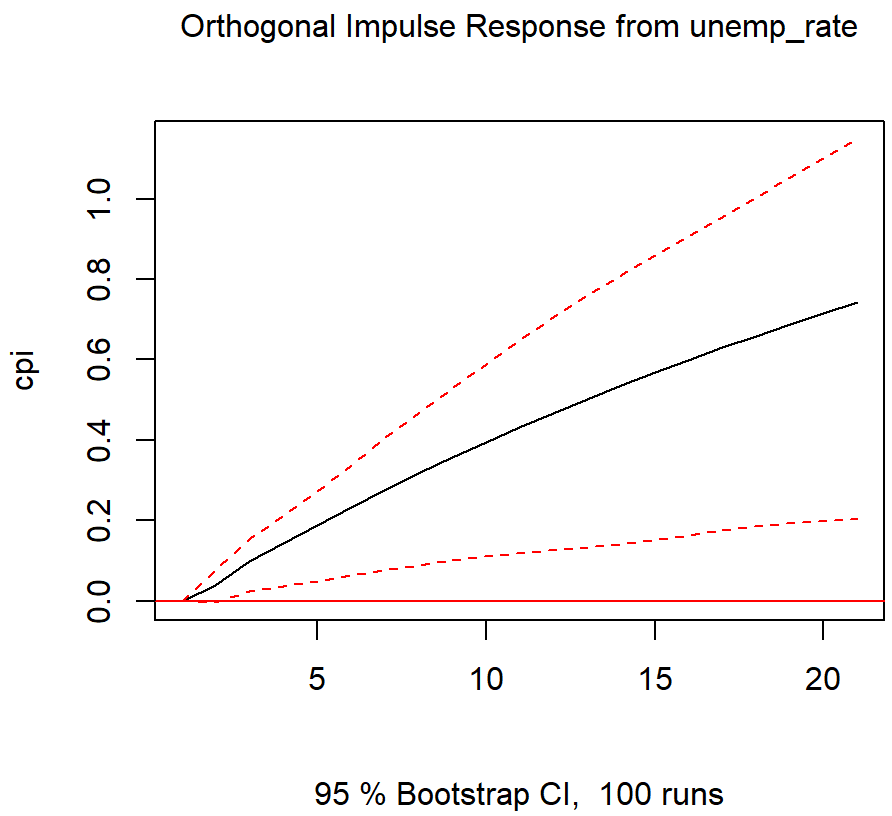
3.8 Impulse Response Analysis

Orthogonal impulse response functions (Cholesky decomposition) are computed over a 20-period horizon with 95% bootstrap confidence intervals. The responses below describe the effect of a one-standard-deviation shock to each variable on the CPI level.


Orthogonal Impulse Response from cpi
cpi
0.2
0.4
0.6
0.8
5
10
15
20
95 % Bootstrap CI, 100 runs


Orthogonal Impulse Response from ind_pro
cpi
0.0
-0.5
-1.0
5
10
15
20
95 % Bootstrap CI, 100 runs


Orthogonal Impulse Response from retail_sales
cpi
0.0
0.4
0.8
1.2
5
10
15
20
95 % Bootstrap CI, 100 runs



CPI own-shock: A one-standard-deviation shock to CPI itself produces a positive response that peaks at approximately period 3 and then gradually declines, but remains persistently above zero through the 20-month horizon. This reflects the strong inflation inertia captured by both the ECT and the significant lagged CPI coefficient.

Supply shock (Industrial Production): A positive shock to industrial production growth initially produces a response near zero and trends mildly negative over the horizon. This dampening effect is consistent with supply expansions increasing productive capacity and moderating price pressures, though the effect is small and slow to materialize.

Demand shock (Retail Sales): A positive shock to retail sales growth produces a response that rises nearly linearly over the horizon, the strongest and most persistent external effect on CPI in the system. This pattern is consistent with demand-pull inflation dynamics and aligns with retail sales being the dominant long-run driver identified in the FEVD.

Labor market shock (Unemployment Rate): A positive shock to the unemployment rate produces a steadily increasing positive response in CPI. While counterintuitive in the short run, as a rise in unemployment might be expected to reduce inflation via the Phillips curve, this long-run positive effect likely reflects that unemployment increases often co-occur with stagflationary periods and that equilibrium adjustment operates through cointegrating relationships rather than contemporaneous effects.

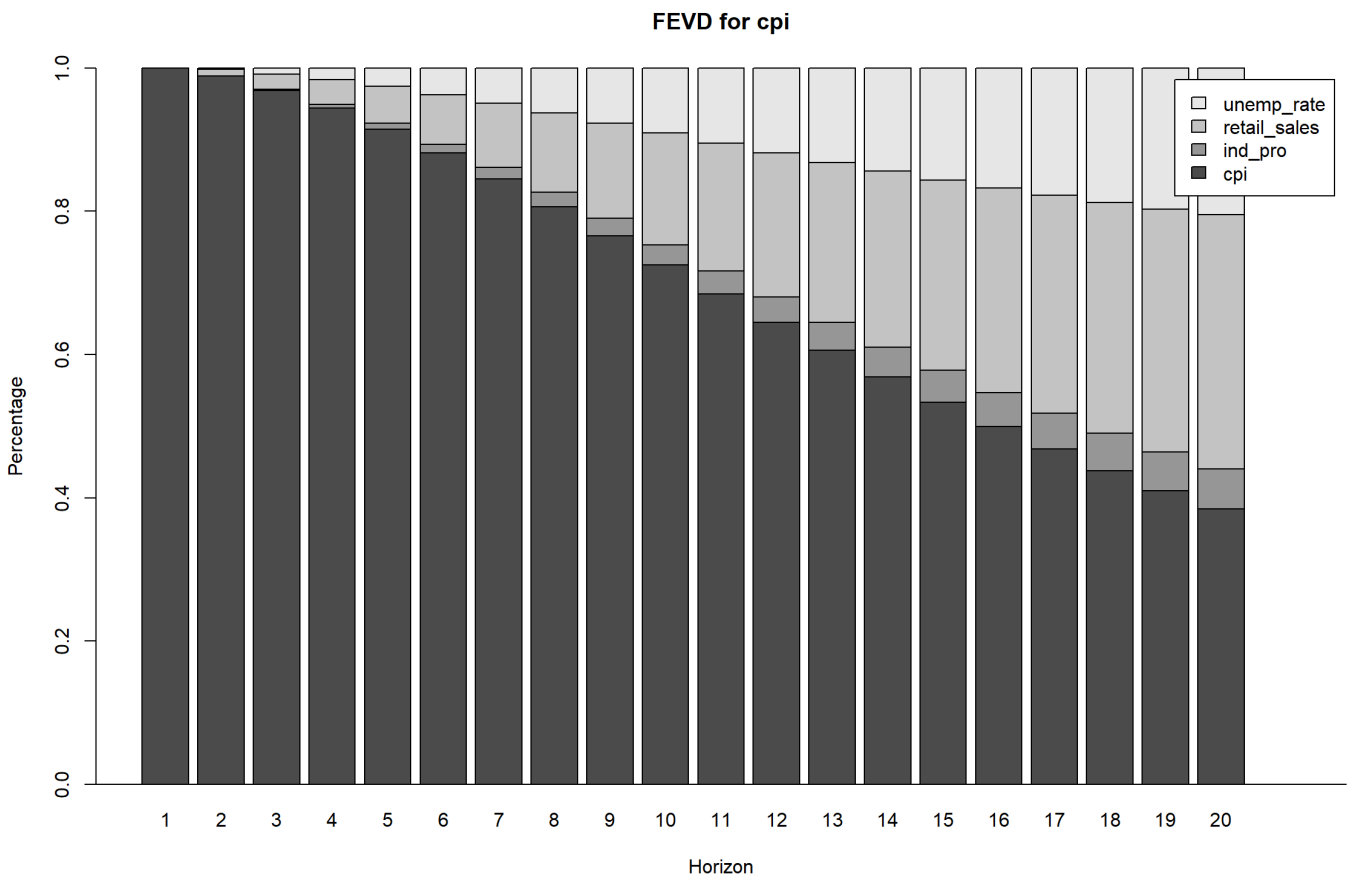
In a VAR on stationary differences, all impulse responses are mean-reverting and converge to zero within roughly 8–10 months. The VECM IRFs, in contrast, exhibit persistent non-zero long-run responses, a direct

consequence of the cointegrated structure and the error correction mechanism. This persistence reflects the genuine long-run dynamics that the VECM is designed to capture.

3.9 Forecast Error Variance Decomposition

The FEVD quantifies the contribution of each variable's shock to the forecast error variance of CPI at different horizons.

Horizon	CPI (own)	Retail Sales	Unemployment	Industrial Production
1 month	~100%	~0%	~0%	~0%
6 months	declining	rising	rising	rising
20 months	38.4%	35.5%	20.5%	5.6%



At short horizons, nearly all forecast error variance in CPI is attributable to its own shocks, a pattern of strong inflation inertia consistent with the significant lagged CPI coefficient in the estimated equation. As the horizon extends, however, the contributions of external shocks grow substantially.

By horizon 20:

- **Retail sales accounts for 35.5%** of CPI forecast error variance, identifying consumer demand as the dominant long-run driver of inflation
- **Unemployment accounts for 20.5%**, confirming that labor market conditions have meaningful long-run inflationary effects despite being insignificant contemporaneously
- **Industrial production accounts for 5.6%**, consistent with supply-side effects being real but secondary
- **CPI own-shock accounts for 38.4%**, reflecting that inflation retains substantial self-driven momentum even over a 20-month horizon

These results differ sharply from the VAR, where the CPI own-shock explained 95.68% of variance at the 24-month horizon, with all external variables contributing less than 5% combined. The VAR on differences, by ignoring the cointegrated structure, severely underestimates the role of demand and labor market shocks in driving inflation dynamics.

3.10 Economic Interpretation

The VECM results yield four substantive economic insights.

1. **Inflation has strong short-run momentum.** The lagged CPI coefficient of 0.317 and the persistent own-shock IRF confirm that month-to-month inflation is substantially driven by its recent history, consistent with inflation expectations and pricing inertia.

2. **Consumer demand is the dominant long-run inflation driver.** Retail sales, the proxy for aggregate demand, explains 35.5% of CPI forecast error variance at 20 months and produces the strongest IRF among external variables. This supports demand-pull inflation theory, as sustained increases in consumer spending translate into persistent upward pressure on prices.

3. **Labor market shocks operate through long-run equilibrium adjustment.** Despite being statistically insignificant in the contemporaneous CPI equation, unemployment contributes 20.5% of long-run CPI variance. This suggests that unemployment affects inflation primarily through the cointegrating relationships, a long-run equilibrium channel, rather than through short-run dynamics.

4. **Supply-side shocks have a modest dampening long-run effect.** Industrial production shocks produce slightly negative long-run CPI responses and explain only 5.6% of long-run variance, suggesting that supply expansions are real but secondary inflation determinants in this system.

The evidence indicates that U.S. inflation is driven primarily by its own persistence and by demand-side conditions, with labor market and supply-side factors playing supporting but meaningful roles over longer horizons.

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4. Conclusion

This paper examines the dynamic relationships between U.S. inflation and three macroeconomic driving forces, industrial production (supply), retail sales (demand), and unemployment (labor market), using monthly FRED data from January 1992 through October 2024.

The analysis begins with the VAR framework as a natural starting point for multivariate macroeconomic modeling. ADF testing confirms that all four series are integrated of order one, I(1), and the Johansen trace test identifies three cointegrating relationships among them. These results indicate that a VAR on first differences is misspecified and motivate the selection of a VECM with $r = 3$ as the preferred model. The VECM preserves the long-run equilibrium information that differencing would discard and introduces error correction terms that capture the speed at which the system adjusts back toward its long-run cointegrating relationships.

The key empirical findings are:

- The error correction term in the CPI equation is negative and highly significant (ect1 = −0.030, $p < 0.01$), confirming that CPI corrects toward long-run equilibrium at approximately 3% per month when deviations occur
- Retail sales is the dominant external driver of long-run inflation, explaining 35.5% of CPI forecast error variance at the 20-month horizon, a finding that is not visible in a VAR framework
- Unemployment contributes 20.5% of long-run CPI variance, with its effects operating primarily through long-run equilibrium adjustment rather than contemporaneous dynamics
- Industrial production has a mild dampening effect on inflation, consistent with supply expansions moderating price pressures
- Across all IRFs, the VECM produces persistent long-run responses, in contrast to the short-lived, mean-reverting responses found in the VAR analysis

Standard diagnostic tests reveal violations of serial independence, homoscedasticity, and normality, attributable primarily to extreme observations during the COVID-19 period (2020–2022). These violations do not invalidate the cointegrating structure or error correction dynamics, but confidence intervals should be interpreted with caution. Extensions that could address these issues include incorporating a structural break dummy for 2020–2022, restricting the sample to the post-2010 period, or employing a regime-switching VECM.

The VECM framework substantially enriches the understanding of U.S. inflation dynamics relative to the VAR baseline. By accounting for the long-run cointegrated structure of the macroeconomic system, the model reveals that inflation is not simply self-driven but is meaningfully shaped by demand conditions and labor market dynamics over medium and long horizons, dynamics that the VAR on differences would not surface.